



Comparing Scores

Standard scores can be used to compare normal distributions.

Comparing z-Scores from Two Distributions

Black tea has been studied for its medicinal properties and for the amount of caffeine it contains (which can vary depending on the type of tea used and the amount of time brewed).

The amount of caffeine in single servings of tea is normally distributed. For a particular comparison, single servings of two different Ceylon tea mixtures are compared.

	Mixture A	Mixture B
Mean	$\mu = 55 \text{ mg}$	$\mu = 52 \text{ mg}$
Standard deviation	$\sigma = 4.1 \text{ mg}$	$\sigma = 3.3 \text{ mg}$

► **Think About It** Ceylon tea is a type of black tea grown in Sri Lanka.

Suppose you are given a serving of Ceylon tea that contains 50 mg of caffeine. Which mixture is the tea more likely to come from?

To determine whether the serving with 50 mg of caffeine is more likely to come from Mixture A or Mixture B, you can determine the z -score for each mixture.

Mixture A	Mixture B
$z = \frac{50 - 55}{4.1}$ ≈ -1.22	$z = \frac{50 - 52}{3.3}$ ≈ -0.61

► **Remember** $z = \frac{x - \mu}{\sigma}$

For a normal distribution, any value that is closer to the mean will be a more likely outcome than any value farther away from the mean. Since -0.61 is closer to 0 than -1.22 , Mixture B is more likely to produce a serving of Ceylon tea with 50 mg of caffeine.

► **Think About It** The z -score of the mean is 0.00 because it is zero standard deviations away from the mean.

Paulo is doing his own research about the caffeine content of these two Ceylon tea mixtures. The brew he makes with Mixture A has 57 mg of caffeine in a single serving. The brew he makes with Mixture B has 49 mg of caffeine in a single serving. Which result is more unusual?

To determine whether the serving with 57 mg of caffeine from Mixture A is more unusual than the serving with 49 mg of caffeine from Mixture B, you can determine the z -score for each mixture.

Mixture A	Mixture B
$z = \frac{57 - 55}{4.1}$ ≈ 0.49	$z = \frac{49 - 52}{3.3}$ ≈ -0.91

The z -scores for each mixture have different signs (one positive and one negative). However, we are only interested in knowing which z -score is farthest from 0 .

Since Mixture A's z -score of 0.49 is closer to 0 than Mixture B's z -score of -0.91 , the serving with 49 mg of caffeine from Mixture B is more unusual than the serving with 57 mg of caffeine from Mixture A.

Interpreting a z-Score in Two Distributions

Two normally distributed data sets can also be compared for the same z-score. For instance, in looking at the two mixtures of Ceylon tea, which mixture has more caffeine for a z-score of -3.00 ?

The z-score is -3.00 , and you know the mean and standard deviation for each mixture. Use the appropriate formula to determine the actual amount of caffeine in a serving from each mixture.

Mixture A	Mixture B
$x = -3 \cdot 4.1 + 55$ $= 42.7$	$x = -3 \cdot 3.3 + 52$ $= 42.1$

At a distance of three standard deviations below the mean, the two mixtures have about the same amounts of caffeine. However, Mixture A has more caffeine compared to Mixture B.

► **Remember** $x = z \cdot \sigma + \mu$